DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. Branch: Electronics & Telecom./EXTC (Sandwich) Semester: IV

Subject Code & Name: BTETPE405A/ BTEXPE405A Numerical Method & Computer Programming

Max Marks: 60 **Date:27/08/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	**An approximate** value of is given by $X1=\frac{22}{7}=3.142857$ and its true value is X=3.1415926 find the absolute and relative errors.	**CO-1**	**6**
B)	**Three approximate** values of the number $\frac{1}{3}$ are given as 0.30, 0.33, 0.34. which of these three is the best approximation?	**CO-1**	**6**
C)	**Evaluate** the sum $S=\sqrt{3}+\sqrt{5}+\sqrt{7}$ to 4 significant digits and find its percentage error?	**CO-1**	**6**
Q.2	**Solve Any Two of the following.**		
A)	Find a root of an equation $f(x)=x^3-3$ using Bisection method	**CO-2**	**6**
B)	Find real root of the equation $x=e^{-x}$ using the Newtons Rapson method	**CO-2**	**6**
C)	Find a root of an equation $f(x)=x^3-x-1$ using False Position method (Regula false method)	**CO-2**	**6**
Q. 3	**Solve Any Two of the following.**		
A)	Find the polynomial f (x) by using Lagrange's formula and hence find f (3) for	**CO-3**	**6**

x	0	1	2	5
F(x)	2	3	12	147

B)	From the following table, estimate the number of students who obtained marks between 40 and 45	**CO-3**	**6**

marks	40—50	50—60	60—0	30—40	30—40
No of students	31	42	51	35	31

C)	Using Gauss backward difference formula, find y (8) from the following		**6**

table

x	0	5	10	15	20	25
y	7	11	14	18	24	32

CO-3

Q.4 Solve Any Two of the following.

A) Value of f(x) in the interval [0,4] are given CO-1 6

x	0	1	2	3	4
F(x)	3	10	21	36	55

Using Simpson's 1/3 rule with the step size of 1. The value of $\int_0^4 f(x)dx$ =?

B) The value of solution f(x) at 5 discreate points are given using trapezoidal rule with step size of 0.1. find all the value of $\int_0^{0.4} f(x)dx$ =? CO-1 6

x	0	0.1	0.2	0.3	0.4
F(x)	0	10	40	90	160

C) Determine the value of y when x=0.1 given that y (0) =1 and $\frac{dy}{dx} = x^2 + y$ h=0.05 using modified Euler's method CO-1 6

Q. 5 Solve Any Two of the following.

A) Explain Basic concept of OOPS CO-6 6

B) Explain data types in c++ CO-6 6

C) What is the basic structure of c++ program? Explain with example. CO-7 6

***** End *****